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REQUIREMENTS, PRIORITISATION AND TRACEABILITY

B2B Web Platform for Wholesale Order Management and Optimisation of Agricultural Product Distribution Flows

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**By signing the cover page of this report, the student confirms their active involvement in both the development of the project and the writing of the report.*

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LIST OF ABBREVIATIONS

API – Application Programming Interface
FR – Functional Requirement
MoSCoW – Must have, Should have, Could have, Won't have
NFR – Non-Functional Requirement
SMART – Specific, Measurable, Achievable, Relevant, Time-bound
SUS – System Usability Scale
TC – Test Case (automated API test)
UC – Use Case

1 INTRODUCTION

This document gathers the requirements of AgroHub, a B2B web platform for wholesale order management and the optimisation of agricultural product distribution flows. It starts from the purpose of the project and its objectives, lists all the functional requirements by module and user role, associates measurable metrics with the non-functional requirements, prioritises everything with the MoSCoW method and links the requirements to the objectives, the design, the implementation and the tests through a traceability matrix.

The purpose of the project is to let Moldovan farmers and wholesale buyers agree and follow wholesale deals in minutes instead of hours of phone calls, by keeping offers, real stock, orders and the conversation about them in one place, between verified participants. Table 1.1 lists the goals and the two SMART objectives that the requirements must serve; the identifiers are used in the traceability matrix in Chapter 5.

Table 1.1 – Goals and objectives of the project

ID	Goal or objective
G1	Shorten the time needed by producers and buyers to find each other and agree on a deal
G2	Keep a single, shared record of stock, orders and agreements
G3	Build trust between partners who do not know each other
G4	Keep the platform free and easy to use for small producers
O1	SMART: during the pilot (1 February – 31 May 2027), the median time between placing an order and the seller's decision is under 4 hours, on at least 30 orders
O2	SMART: during the pilot, an average SUS score of at least 70 among at least 15 users, and 80% of new sellers publish a first listing in under 3 minutes

The requirements were formulated following ISO/IEC/IEEE 29148 [1], the non-functional ones were classified with the quality model of ISO/IEC 25010:2023 [2], and the detailed use cases and user stories are given in the software requirements specification of the project.

2 FUNCTIONAL REQUIREMENTS

The functional requirements describe what the system does for each of its users. They are organised in five modules, each serving one or more roles: the guest (a visitor who has not signed in), the seller (a producer), the distributor (a wholesale buyer) and the administrator of the platform. The column “Priority” gives the MoSCoW category explained in Chapter 4.

The first module, accounts and access, controls who can enter the platform and with which role. Its requirements are listed in Table 2.1.

Table 2.1 – Functional requirements: accounts and access

ID	Requirement	Role	Priority
FR-01	Registration. The system allows a guest to create a seller or distributor account with name, e-mail, password and, optionally, phone and region	Guest	Must
FR-02	E-mail verification. The system sends a six-digit code to the e-mail address and activates the account only after the correct code is entered	Guest	Must
FR-03	Authentication. The system allows signing in with e-mail and password	All	Must
FR-04	Google sign-in. The system allows signing in or registering with a Google account	All	Should
FR-05	Password recovery. The system allows setting a new password with a code sent by e-mail	All	Must
FR-06	Account approval. The system approves verified accounts automatically or leaves them to the administrator, who can approve, reject or block any account	Administrator	Must

Signing in with Google (FR-04) is a convenience rather than a necessity, which is why it is the only requirement of the module that is not a Must.

The second module covers the catalogue, through which buyers find offers, and the listings, through which sellers publish them. Table 2.2 lists its requirements.

Table 2.2 – Functional requirements: catalogue and listings

ID	Requirement	Role	Priority
FR-07	Catalogue search. The system displays the active listings and allows searching by product or variety and filtering by region	All	Must
FR-08	Listing details. The system displays the quantity, price per kilogram, region, harvest date, delivery terms, contacts and rating of the seller	All	Must
FR-09	Listing management. The system allows a seller to create, edit, pause and delete listings; a new listing requires a photo	Seller	Must
FR-10	Stock adjustment. The system allows a seller to correct the stock of a listing with a stated reason and records every change	Seller	Should
FR-11	Automatic listing check. The system checks every new or edited listing with rules and an AI service and leaves doubtful listings to the administrator, with the reasons	System	Must

The automatic check of listings (FR-11) lets trusted sellers publish in seconds while keeping spam and unrealistic offers out of the catalogue.

The orders module, in Table 2.3, implements the life cycle of a deal: pending, confirmed, rejected, cancelled and delivered.

Table 2.3 – Functional requirements: orders

ID	Requirement	Role	Priority
FR-12	Placing an order. The system allows a distributor to order a quantity no larger than the available stock, with an optional message; the price is fixed at the moment of ordering	Distributor	Must
FR-13	Accepting or rejecting. The system allows the seller to accept or reject a pending order; on acceptance the stock decreases and a listing with zero stock is paused	Seller	Must
FR-14	Editing or cancelling. The system allows the distributor to edit and resend an order that is pending, rejected or cancelled, and to cancel an order that is not yet delivered; the stock of an accepted order is restored	Distributor	Must
FR-15	Delivery confirmation. The system allows the seller to mark an accepted order as delivered, after which it can no longer be cancelled	Seller	Must
FR-16	Personal dashboard. The system shows each user the statistics and the history of their listings and orders, and lets them hide old orders from their own history	Seller, distributor	Must

All the requirements of this module are Must, because without them the platform would not replace the oral agreements by phone, which is the core of its value.

Table 2.4 contains the requirements for communication and reviews, and Table 2.5 those of the administrator.

Table 2.4 – Functional requirements: communication and reviews

ID	Requirement	Role	Priority
FR-17	Order conversation. The system provides a conversation for every order and shows the number of unread messages	Seller, distributor	Must
FR-18	Notifications. The system creates an in-app notification and sends an e-mail for every change of an order; for a new message it sends an e-mail, unless the recipient has read the conversation in the last 30 seconds	Seller, distributor, administrator	Must
FR-19	Reviews. The system allows both parties of an accepted or delivered order to give a rating from 1 to 5 with a comment, and the reviewed user to reply; the average rating is public	Seller, distributor	Must

Notifications (FR-18) are directly linked to objective O1, since the seller can answer an order quickly only if the seller learns about it at once.

The administration module gives the platform staff the tools to keep the catalogue and the reviews trustworthy.

Table 2.5 – Functional requirements: administration

ID	Requirement	Role	Priority
FR-20	Moderation of listings and reviews. The system allows the administrator to approve or reject listings, to request a new AI check and to delete reviews or replies	Administrator	Must
FR-21	Moderation settings. The system allows the administrator to change the automatic approval rules: thresholds for trusted sellers, price limits, banned words and the AI check	Administrator	Should
FR-22	Daily summary. The system sends the administrators a daily e-mail summary of the moderation events at a configured hour, Moldova time	Administrator	Could

In total, the system has 22 functional requirements. Some functions were consciously left out of this version; they are listed as Won't have in Chapter 4, so that their absence is a documented decision and not an omission.

3 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements describe the qualities of the system. Each one is classified according to ISO/IEC 25010:2023 and has a measurable metric, so that it can be verified objectively. Table 3.1 contains the requirements for performance and reliability.

Table 3.1 – Non-functional requirements: performance and reliability

ID	Category (ISO/IEC 25010)	Requirement and metric	Priority
NFR-01	Performance efficiency (time behaviour)	The catalogue and the order operations respond quickly. 95% of the requests for the catalogue, placing an order and changing its status are answered in under 2 seconds, measured on 200 requests with 1,000 active listings	Should
NFR-02	Performance efficiency (capacity)	The system serves many users at the same time. With 50 simultaneous virtual users for 5 minutes, fewer than 1% of the requests fail and 95% are answered in under 3 seconds	Should
NFR-03	Reliability (availability)	The public version of the platform is available. Monthly availability of at least 99%, measured by an external monitor that calls the health endpoint every 5 minutes	Should
NFR-04	Reliability (faultlessness)	The stock never becomes inconsistent. In 100% of the runs of the test with two simultaneous acceptances, exactly one succeeds and the stock decreases once; the stock can never be negative	Must
NFR-05	Reliability (fault tolerance, recoverability)	An outage of the AI service does not block the publication of listings. If the AI service does not answer within 45 seconds, the listing goes to manual moderation and is rechecked automatically every 15 minutes, up to 12 times; no listing is lost	Should

The availability target of 99% is lower than the 99.9% often cited for commercial services, because the first version runs on free hosting plans; it can be raised when the platform moves to paid infrastructure.

Table 3.2 contains the requirements for security, ease of use, adaptability and maintainability.

Table 3.2 – Non-functional requirements: security, usability, flexibility and maintainability

ID	Category (ISO/IEC 25010)	Requirement and metric	Priority
NFR-06	Security (authenticity, confidentiality)	Passwords and codes cannot be read or guessed. The database contains 0 passwords or codes in clear text (bcrypt and SHA-256 only); codes expire after 15 minutes, the 6th wrong attempt is refused; all traffic uses HTTPS	Must
NFR-07	Security (resistance)	The system resists repeated attempts and abuse. The 21st sign-in attempt and the 6th e-mail request from the same address within 15 minutes receive HTTP 429	Must
NFR-08	Security (confidentiality, integrity)	Each user reaches only the data and actions of their role. 100% of the protected requests return 401 without a token and 403 for a wrong role or a blocked account; only name, e-mail, phone and region are stored as personal data	Must
NFR-09	Interaction capability (operability, learnability)	Frequent operations are fast to perform. In a test with 3 new users, each publishes a first listing in at most 3 minutes and places an order in at most 5 actions from the catalogue, without help	Should
NFR-10	Flexibility (adaptability)	The platform works on the devices of the target users. The flows F1–F3 run without errors in the last two versions of Chrome, Firefox, Safari and Edge and on screens from 360 pixels wide, without horizontal scrolling	Must
NFR-11	Maintainability (testability)	Changes can be verified quickly. Every mandatory functional requirement is covered by at least one automated test; the 19 API tests run on every push and finish in under 5 minutes	Should

Requirement NFR-09 uses the same metrics as objective O2, and NFR-01 supports objective O1, so the success of the project and the quality of the system are measured with the same numbers.

4 PRIORITISATION OF THE REQUIREMENTS

The requirements were prioritised with the MoSCoW method, which divides them into four categories [3]: Must have (without it the system has no value or cannot be used safely), Should have (important, but the system works without it), Could have (useful if time allows) and Won't have (consciously excluded from this version). The criteria were the contribution to the objectives O1 and O2 and to the key flows of the users, the risk of leaving a requirement out, and the effort compared to the time available.

Table 4.1 shows how the requirements are distributed among the categories.

Table 4.1 – Distribution of the requirements by MoSCoW category

Category	Functional	Non-functional	Examples
Must have	18	5	Placing and accepting orders, stock linked to orders, access control
Should have	3	6	Google sign-in, manual stock adjustment, response time
Could have	1	0	Daily summary for the administrators
Won't have	5	0	Online payments, route planning (Table 4.2)

Most functional requirements are Must, which is normal for a minimum viable product: each of them covers a step of the key flows. Among the non-functional requirements, those that protect data and stock are Must, while the performance targets are Should, because they are verified under load only in the pilot phase.

The functions excluded from this version are listed in Table 4.2, together with the reason.

Table 4.2 – Won't have requirements of the current version

ID	Function	Reason for exclusion
W-01	Online payments between buyers and sellers	Requires a payment provider and contracts; outside the budget and the scope
W-02	Issuing fiscal documents and electronic invoices	Depends on national e-invoicing systems; planned for a later version
W-03	Transport planning and route optimisation	Physical transport stays with the parties; the project optimises the information flow
W-04	Native mobile applications	The responsive web interface covers phones without installation
W-05	Interface in Russian and English	The target users of the first version work in Romanian

These functions are not rejected for good. They are the first candidates for the next versions, once the pilot shows how the platform is actually used.

5 TRACEABILITY MATRIX

The traceability matrix shows that every requirement serves the objectives of the project and that every requirement is designed, implemented and tested. It is read in two directions: from an objective to the requirements that achieve it (Table 5.1), and from each requirement to its use case, its implementation and its tests (Tables 5.2–5.4).

Table 5.1 – Objectives and goals traced to requirements

Objective	Functional requirements	Non-functional requirements	Verification
O1. Decision on an order in under 4 hours	FR-12, FR-13, FR-17, FR-18	NFR-01, NFR-03	Pilot timestamps; TC08, TC10, TC14
O2. SUS ≥ 70; first listing in under 3 minutes	FR-04, FR-09, FR-11, FR-16	NFR-09, NFR-10	SUS questionnaire; timed first use
G1. Less time to find a partner	FR-07, FR-08, FR-11, FR-12, FR-13, FR-17, FR-18	NFR-01, NFR-02, NFR-05	TC05, TC08, TC10
G2. One shared record	FR-10, FR-12 to FR-17	NFR-04, NFR-11	TC10 to TC16
G3. Trust between partners	FR-02, FR-03, FR-06, FR-11, FR-15, FR-19 to FR-22	NFR-05 to NFR-08	TC01, TC05, TC06, TC17, TC19
G4. Free and easy to use	FR-01, FR-04, FR-05, FR-09	NFR-09, NFR-10	Test with new users

Every goal and objective is supported by at least two functional requirements and one non-functional requirement, and no requirement is left without a goal.

Table 5.2 traces the non-functional requirements to the goals and to the method used to verify them.

Table 5.2 – Traceability of the non-functional requirements

NFR	Priority	Goals and objectives	Verification
NFR-01	Should	G1, O1	Load test, 200 requests
NFR-02	Should	G1	Load test, 50 virtual users
NFR-03	Should	G1, O1	External uptime monitor
NFR-04	Must	G2	TC11, TC12
NFR-05	Should	G1, G3	AI outage simulation
NFR-06	Must	G3	Database inspection
NFR-07	Must	G3	Repeated requests test
NFR-08	Must	G3	TC03, TC04, TC09, TC19
NFR-09	Should	G4, O2	Test with 3 new users, SUS
NFR-10	Must	G4, O2	Manual test on 4 browsers
NFR-11	Should	G2	Automated tests on every push

Three non-functional requirements (NFR-04, NFR-08 and NFR-11) are already verified by the automated tests; the load, availability and usability targets will be measured before and during the pilot.

Tables 5.3 and 5.4 trace each functional requirement to the use case of the specification, the API endpoints that implement it and the tests that verify it. “Manual” marks the requirements that are verified by a manual test scenario instead of an automated one.

Table 5.3 – Traceability of the functional requirements FR-01 to FR-11

FR	Priority	Goals and objectives	Use case	Implementation (API, /api)	Tests
FR-01	Must	G4	UC02	POST /auth/register	TC01
FR-02	Must	G3	UC03	POST /auth/verify-email, /auth/resend-code	TC01
FR-03	Must	G3	UC04	POST /auth/login	TC01, TC02
FR-04	Should	G4, O2	UC04a	POST /auth/google	Manual
FR-05	Must	G4	UC04	POST /auth/forgot-password, /auth/reset-password	Manual
FR-06	Must	G3	UC14	PATCH /admin/users/:id/status	TC01, TC19
FR-07	Must	G1	UC01	GET /listings	TC05
FR-08	Must	G1, G3	UC01	GET /listings, /listings/:id/image	Manual
FR-09	Must	G1, G4, O2	UC05	POST, PATCH, DELETE /listings; PATCH /listings/:id/status	TC04, TC05
FR-10	Should	G2	UC05	PATCH /listings/:id/stock	Manual
FR-11	Must	G1, G3, O2	UC06, UC15	Moderation module (rules and AI check)	TC05, TC06

All the requirements of accounts, catalogue and listings are implemented. Google sign-in and password recovery depend on an external account or an e-mail inbox, while the listing details and the stock adjustment are checked through the interface, so these four are tested manually.

Table 5.4 continues the matrix with the orders, communication, reviews and administration.

Table 5.4 – Traceability of the functional requirements FR-12 to FR-22

FR	Priority	Goals and objectives	Use case	Implementation (API, /api)	Tests
FR-12	Must	G1, G2, O1	UC07	POST /orders	TC07, TC08
FR-13	Must	G1, G2, O1	UC09	PATCH /orders/:id/status	TC09–TC12
FR-14	Must	G2	UC08	PATCH /orders/:id, /orders/:id/cancel	TC15, TC16
FR-15	Must	G2, G3	UC10	PATCH /orders/:id/complete	TC13, TC14
FR-16	Must	G2, O2	UC13	GET /dashboard/me	Manual
FR-17	Must	G1, G2, O1	UC11	GET, POST /orders/:id/messages	Manual
FR-18	Must	G1, O1	UC13	GET /notifications; notification module	TC08, TC14, TC18
FR-19	Must	G3	UC12	POST /orders/:id/review, PUT /reviews/:id/reply	TC17
FR-20	Must	G3	UC15	PATCH /admin/listings/:id/status, POST /admin/listings/:id/recheck, DELETE /admin/reviews/:id	TC05
FR-21	Should	G3	UC16	GET, PUT /admin/moderation/settings	Manual
FR-22	Could	G3	UC16	Daily digest scheduler	Manual

Every Must requirement of the orders module is covered by at least one automated test, including the simultaneous acceptance of the same order (TC12).

The matrix will be completed during the next stages: the UML models of the design phase will be added as a new column, and the results of the load, usability and pilot tests will replace the planned verification methods.

6 CONCLUSIONS

The platform is described by 22 functional requirements, organised in five modules and four roles, and by 11 non-functional requirements with measurable metrics. The MoSCoW prioritisation separates 23 Must requirements, which form the minimum viable product, from those that can wait, and five functions are consciously excluded from this version.

The traceability matrix links the two SMART objectives and the four goals of the project to the requirements, the use cases, the API endpoints and the tests. Most functional requirements are already verified by automated tests, and the rest by manual scenarios, so every requirement can be followed from the purpose of the project to its verification.

REFERENCES

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[3] D. Clegg and R. Barker, Case Method Fast-Track: A RAD Approach. Wokingham, UK: Addison-Wesley, 1994.